SAP Blueprint Project

Capstone Project Documentation

Shabnam Praveen

Roll No: 2329209 | Batch: 2023–27 | SAP BTP Developer, KIIT

I. PROBLEM STATEMENT

In modern enterprises, managing procurement, sales, and finance as isolated processes leads to data silos, reconciliation errors, delayed reporting, and poor inventory visibility. Without an integrated ERP system, organisations rely on manual handoffs between departments, resulting in operational inefficiencies and inaccurate financial records.

This project addresses the need to design and simulate a fully integrated SAP landscape for a fictitious company. It demonstrates how SAP's core modules—MM, SD, and FI—interact in real business scenarios, providing a blueprint that reflects authentic SAP customisation and configuration logic as practised in industry implementations.

II. SOLUTION OVERVIEW

The solution is a seven-sheet Excel workbook (*SAP_Blueprint_Project*) hosted on OneDrive for live cross-sheet formula resolution. Each sheet corresponds to a distinct SAP functional area or organisational layer, collectively forming a coherent implementation blueprint. The workbook covers the complete Procure-to-Pay, Order-to-Cash, and Record-to-Report cycles.

III. FEATURES AND MODULE DESCRIPTION

A. Company Structure

- Defines SAP org units: Company Code, Plant, Storage Location
- Sales Organisation, Distribution Channel, Division
- Backbone for all transactional modules

B. Procurement — MM (Purchase_Process_MM)

- 5 Purchase Orders: PO001–PO005
- Materials: Laptop, PC, CPU, Mouse, Keyboard
- Vendors: SP Ltd, ABC Suppliers, QC Distributors, KP Agency
- Status: Delivered / Ordered / Invoiced
- Auto Inventory flag via IF formula
- Colour-coded conditional formatting

C. Inventory Sheet

- Tracks live stock levels linked to PO delivery
- Eliminates manual warehouse–finance reconciliation

D. Sales — SD (Sales_SD Sheet)

- 5 Sales Orders: SO001–SO005
- Customers: Srija Enterprises, SKG Traders, Gupta Electronics, Gada Electronics
- Delivery Status: Delivered / Pending
- Billing dropdown: Paid / Not Paid
- Flow: Order → Delivery → Billing

E. Finance — FI (Finance_FI Sheet)

- Records Debit → Credit accounting entries
- Reflects financial impact of MM and SD transactions
- Simulates SAP FI Record-to-Report cycle

F. Dashboard & Process Flow

- Dashboard: Total POs, SOs, Inventory via COUNTA
- Cross-sheet formulas for real-time KPI aggregation
- Process Flow: MM (PR→PO→GR→Invoice)
- SD (Order→Delivery→Billing), FI (Debit→Credit)

IV. SCREENSHOTS

The figures below illustrate the key sheets of the SAP_Blueprint_Project workbook, demonstrating the data structures, formula logic, and visual indicators implemented across the procurement, sales, finance, and dashboard modules.

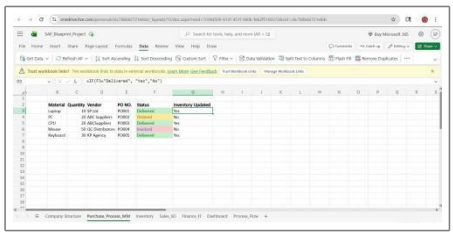

Figure 1. Purchase_Process_MM Sheet — Purchase Orders with vendor, quantity, delivery status and auto-populated Inventory Updated column (IF formula).

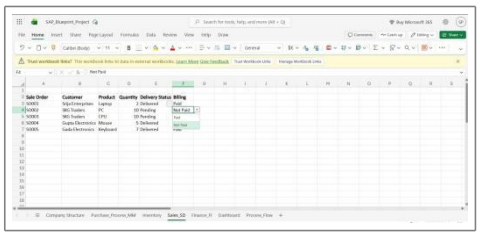

Figure 2. Sales_SD Sheet — Sales Orders showing customer, product, delivery status and billing dropdown validation (Paid / Not Paid).

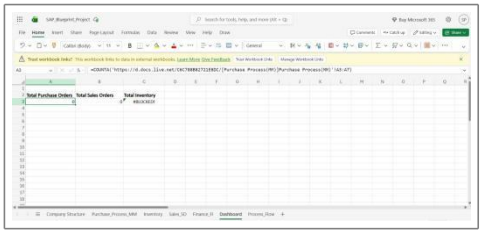

Figure 3. Dashboard Sheet — Live KPI summary: Total Purchase Orders, Total Sales Orders, and Total Inventory aggregated via cross-sheet COUNTA formulas.

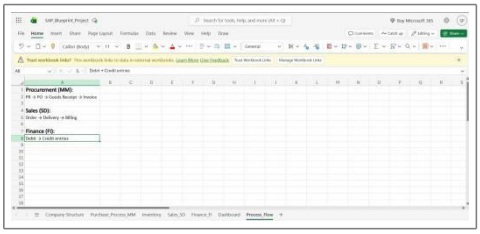

Figure 4. Process_Flow Sheet — End-to-end documented process flows for MM, SD, and FI modules in the SAP blueprint.

V. TECH STACK

Tool / Technology	Application in Project
Microsoft Excel (OneDrive)	Primary simulation platform — sheets, formulas, validation, formatting
SAP MM – Materials Management	Procurement blueprint: PR → PO → Goods Receipt → Invoice
SAP SD – Sales & Distribution	Sales cycle: Order → Delivery → Billing; customer and product management
SAP FI – Financial Accounting	Debit/Credit postings; FI-MM and FI-SD integration simulation
Excel IF Formula	Auto-populates Inventory Updated status based on delivery confirmation
Excel COUNTA Formula	Dashboard KPI aggregation across Purchase_Process_MM, Sales_SD, Inventory sheets
Data Validation (Dropdown)	Controls Delivery Status and Billing fields — prevents invalid data entry
Conditional Formatting	Colour-coded cells for instant visual identification of order and payment states

VI. UNIQUE POINTS

1. End-to-End SAP Integration	All three core SAP modules (MM, SD, FI) are modelled within a single workbook, replicating inter-module posting logic of a live SAP system.
2. Live Dashboard Aggregation	Cross-sheet COUNTA formulas provide real-time KPIs equivalent to SAP reporting transactions (ME2M, VA05, F.01) without any manual refresh.
3. Data Validation Controls	Dropdown lists for Delivery Status and Billing replicate SAP's enumerated status fields, ensuring data integrity and preventing invalid entries.
4. Automated Inventory Flag	IF-based formula auto-updates Inventory Updated on delivery, mirroring the automatic inventory posting triggered by a Goods Receipt (MIGO) in SAP MM.
5. SAP Blueprint Methodology	Follows the standard SAP Blueprint documentation approach : organisational structure, process flows, and transactional data in one artefact.

VII. FUTURE IMPROVEMENTS

The current Excel-based blueprint provides a solid conceptual and functional simulation. The following enhancements are proposed for subsequent iterations of this project:

1. Migration to SAP S/4HANA Sandbox	Execute live transactions via ME21N (Create PO), VA01 (Create SO), and F-02 (GL Posting) to validate configuration against real SAP business logic.
2. SAP HR – Hire-to-Retire Module	Simulate the complete employee lifecycle: recruitment, onboarding, payroll, leave management, and separation within the SAP HCM module.
3. Production Planning (SAP PP)	Add BOM, work centres, and manufacturing orders to connect procurement and inventory layers to a Plan-to-Produce scenario.
4. Power BI Integration	Connect the workbook to Power BI for advanced visualisation, trend analysis, and management-ready drill-through dashboards.
5. Custom ABAP ALV Report	Develop an ABAP program using the ALV framework to programmatically extract and display procurement and sales KPIs.
6. VBA Automation for Inter-Module Postings	Implement macros to auto-trigger FI postings on MM goods receipt and SD billing, replicating SAP's automatic account determination.

VIII. CONCLUSION

This capstone project successfully demonstrates a structured SAP blueprint encompassing the Materials Management, Sales & Distribution, and Financial Accounting modules of the SAP ERP system. The Excel-based simulation provides a tangible artefact that mirrors real SAP behaviour through formulas, data validation, conditional formatting, and cross-sheet integration. The project reinforces the theoretical and practical knowledge gained during the SAP training programme and establishes a strong foundation for live SAP implementation work in professional environments.

End of Document — SAP Blueprint Project Capstone Report